Master's Thesis: Status and Plan

Biofilm growth coupled into the IKM FEM element, following Klempt et al.

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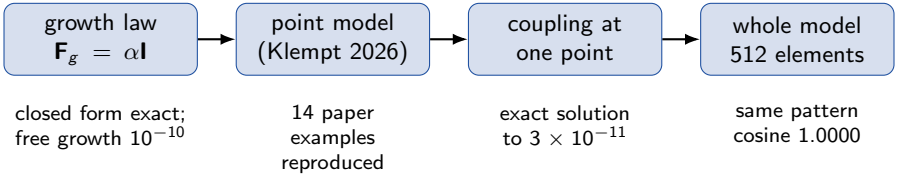
20 October 2026

Since the last meeting: the scope of the FEM part is fixed

	content	reference	coupling
thesis (LUH)			
one species	growth law, Eq. 36	Klempt et al. 2024	field → growth
two species	composition from the point model	Klempt et al. 2026	field → composition (one-way)
continuation (Keio)			
five species	calibrated point model; first runs in Appendix G	none yet	two-way: composition and stress act back

One and two species can be checked against published examples; five species have no reference yet.

Verification, step by step



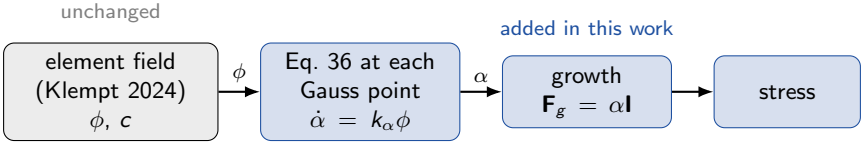
Each step is checked before the next one builds on it.

- point model: two misprints in Klempt et al. 2026 identified (no $+\gamma$ in Eq. 17, no $\frac{1}{2}$ in Eq. 19)
- same point model in Fritsch et al. 2025: their figures reproduced within 0.013

The coupling: what the element does and what is added

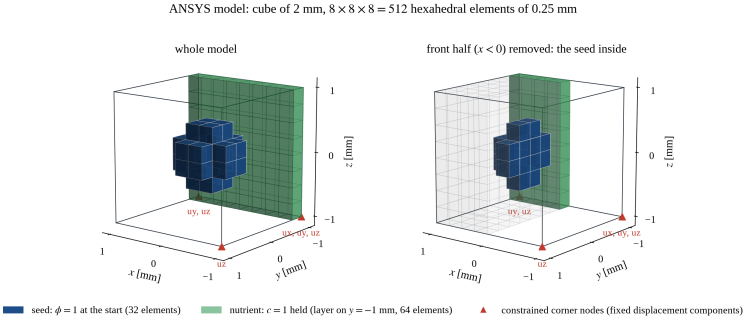
The element solves the biofilm field of Klempt et al. 2024 (Eq. 34):

$$\dot{\phi} - \beta\,\Delta\phi - k_\alpha\,\alpha + |\nabla\phi|\,r\,\frac{c}{k+c}\,\mathbf{n}_\phi \cdot \mathbf{n}_c = 0$$



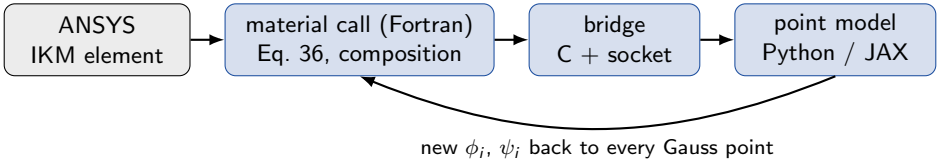
Klempt 2024 Table 2: $k_\alpha = 10^{-3}$, $E = 10$ Pa, $\nu = 0.49$. β, r, k, d, g: the framework's example input, not from the paper.

Model and boundary conditions



- 2 mm cube, 512 elements; only rigid-body motion fixed
- nutrient $c = 1$ on the face $y = -1$ mm; seed of 32 elements with $\phi = 1$
- uniform growth is stress free; stress comes from differences in growth
- used for Checks 1–3 and the two-species runs (ANSYS); the Klempt 2024 reproduction uses the paper's 20 μm cube (Python)

What I implemented



speed single-element run 357 s → **5.6 s**

check every ANSYS call replayed in Python, difference 0

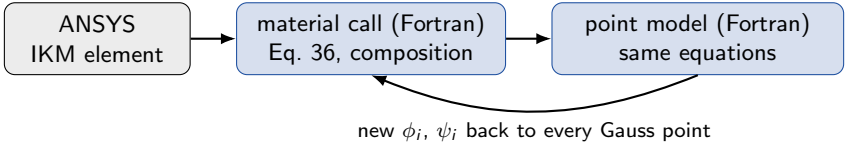
used as it is

- IKM element: field, stiffness
- point-model equations (Klempt et al. 2026)

written in this work

- Eq. 36 and the composition step in the material call
- Fortran → Python bridge, sub-stepping
- point-model solver (JAX): all 14 paper examples
- checks: closed form in ANSYS, call replay, about 440 tests

Second version: the point model in Fortran



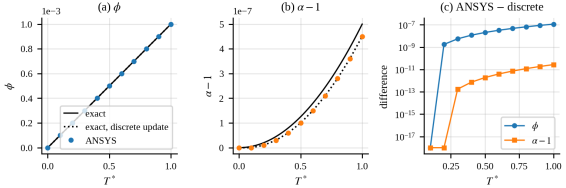
- replaces the bridge and the Python process; the material call is unchanged
- per sub-step: Newton on the 12 unknowns $\phi_1..\phi_5,\ \phi_0,\ \psi_1..\psi_5,\ \gamma$ (6 iterations), exact Jacobian written out, LU with pivoting

check	**result**
against the Python solver (16 states, 4 cases)	largest relative difference 8×10^{-15}
ANSYS, 8^3, two species, one week	stress, nutrient identical; point model $\leq 1.6 \times 10^{-12}$
run time, same model	156 s $\rightarrow$ **36 s**

No server or port: one executable, so runs on a cluster or in parallel need no extra process.

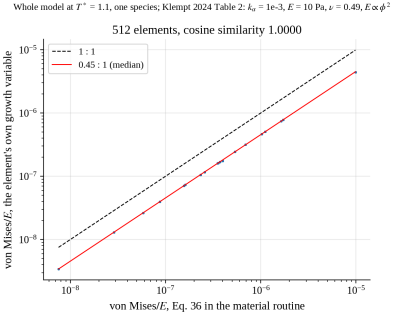
Checks 1 and 2: exact solution and whole model

1. Exact solution



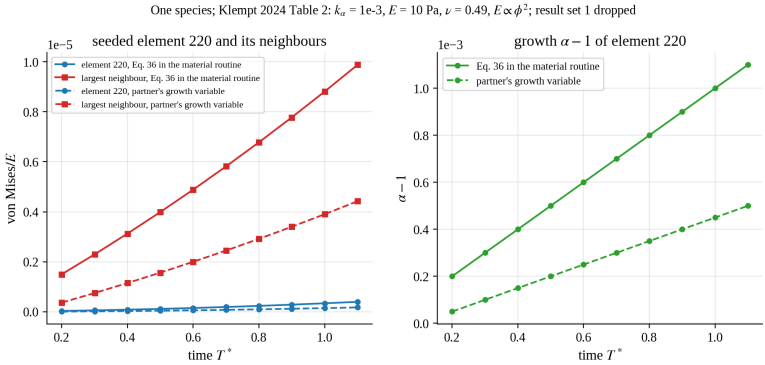
α matches to 3×10^{-11}

2. Whole model, 512 elements



same pattern (cosine 1.0000); factor $0.455 = \frac{1}{2} \cdot \frac{10}{11}$

Check 3: the seeded element and its neighbours



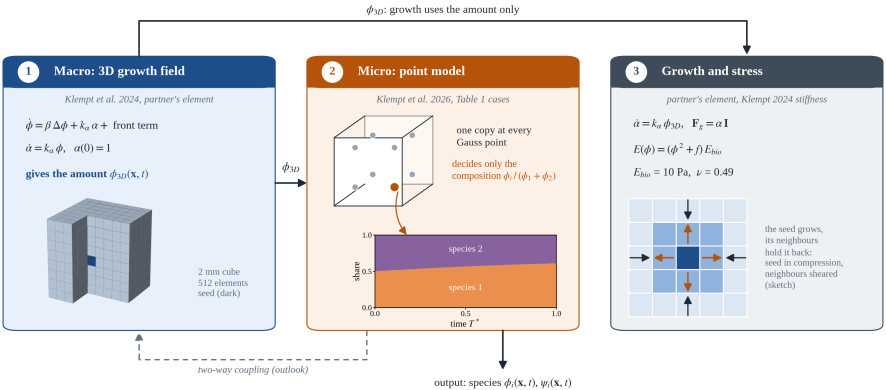
- $\alpha - 1 = 1.1 \times 10^{-3} = k_\alpha \phi t$, as Eq. 36 says
- seed compressed; neighbours carry about 25× its von Mises stress
- von Mises/$E \leq 10^{-5}$: slow growth (stresses scale with E)

Four things found along the way (partner element)

	what I found	effect	what I did
1	growth variable is the mean of both species fields	growth halved with one species (unused field stays at its start value)	asked Oliver whether intended
2	start values must satisfy $\phi_1 + \phi_2 \leq 1$	example input starts both at 1	one species: second field starts at 0
3	first result set not in equilibrium	zero stress while the element sets up its operators	first result set excluded
4	front-growth term inactive (reading slip); $\lvert\Delta\phi\rvert$ where Eq. 34 has $\lvert\nabla\phi\rvert$	seed does not spread	corrected in a test copy: then ϕ exceeds 1

Checks 1–3 do not depend on these four points.

Two species: composition from the point model (Klempt et al. 2026)



Point model run two ways: A) rescaled every step to ϕ_{3D}; B) started when ϕ_{3D} first reaches ϕ_{min}, then independent. Assumed, not from a paper: $s = 0.15$, $\phi_{cap} = 0.9$, $\phi_{min} = 0.01$.

- the field sets the **amount**, the point model the **composition**
- one-way: the composition does not act on the stress

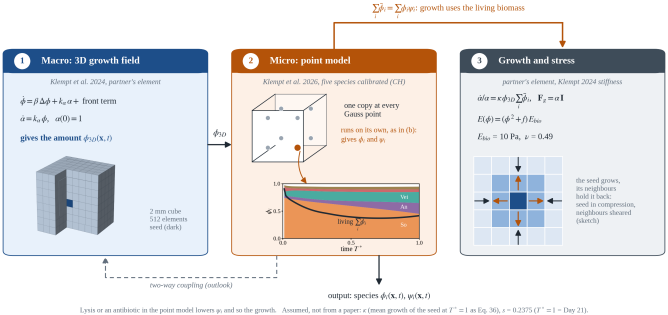
Two species: results in the packed seed (ANSYS, $T^* = 1$)

case (Klempt et al. 2026)	paper	coupled, $\phi_{\mathrm{cap}} = 0.9$	$\phi_{\mathrm{cap}} = 0.85$ / 0.95
3: coexistence	0.60	0.667	0.668 / 0.664
6: one species wins	≈ 0 ($\psi_1 = 0.05$)	0.021 ($\psi_1 = 0.048$)	–

share $\phi_1/(\phi_1 + \phi_2)$; paper: point model alone; parameters $c^* = 100$, $\alpha^* = 10$

- same type of outcome; the ratio shifts because the field sets the amount
- converged in the coupling time step

Added since 5 October: growth from the living biomass



Lysis or an antibiotic in the point model lowers ψ_i and so the growth. Assumed, not from a paper: κ (mean growth of the seed at $T^* = 1$ as Eq. 36), $s = 0.2375$ ($T^* = 1$ = Day 21).

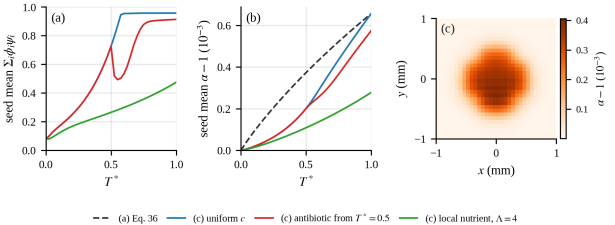
Eq. 36 uses the amount: $\dot{\alpha} = k_\alpha \phi$.
Here: $\dot{\alpha}/\alpha = \kappa\, \phi_{3D} \sum_i \phi_i \psi_i$, so lysis in the point model lowers the growth.

κ chosen so that the mean growth of the seed equals Eq. 36 (an assumption, not from a paper; Appendix E).

seed, $T^* = 1$	Eq. 36	living	diff.
$\alpha - 1$, 8^3	$6.752 \cdot 10^{-4}$	$6.752 \cdot 10^{-4}$	0.0 %
von Mises/E, 8^3	$4.684 \cdot 10^{-5}$	$4.357 \cdot 10^{-5}$	−7.0 %
von Mises/E, 16^3; 24^3			−7.1; −7.0 %
antibiotic vs. none, 8^3	0 %	−32 %	

five species calibrated (CH), $E = 10$ Pa; the antibiotic case with the previous calibration

Growth from the living biomass with synthetic data



Same seed problem; point model: four-species case of Fritsch et al. 2025 with its true parameters (no measured data), own void, start $\phi_i = 0.02$. $T^* = 1$ spans their 1500-step validation run ($s = 0.15$); antibiotic $\alpha^* = 50$ from $T^* = 0.5$ as in their Fig. 18. κ matched to the seed growth of Eq. 36 at $T^* = 1$ (an assumption), or $\kappa = k_\alpha$ of Klempt et al. 2024. Local nutrient: $c^* = 25\,c(x)$, $c = 1$ on one face, $\Lambda^2 = gL^2/d = 16$. (c): $\alpha - 1$ on the mid-plane with the local nutrient.

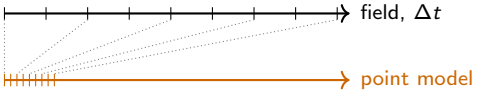
seed, $T^* = 1$	κ matched		$\kappa = k_\alpha$ (nothing fitted)	
	$\alpha - 1$	von Mises	$\alpha - 1$	von Mises
Eq. 36	$6.56 \cdot 10^{-4}$	$6.07 \cdot 10^{-4}$ Pa		
living, uniform c	same	+12 %	−42 %	−35 %
antibiotic	−12 %	−13 %	−12 %	−13 %
local nutrient	−58 %	−72 %	−58 %	−72 %

antibiotic and nutrient rows: change against the uniform-c run of the same κ.

The sign of the stress change follows the time course of the living biomass: here it rises from 0.08 (growth comes late, +12 %), in the calibrated run it falls (−7 %).

Two modelling assumptions, each with a sensitivity study

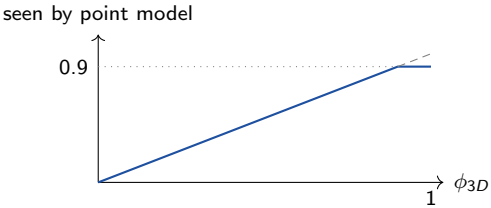
1. time link $s = 0.15$



each step: point model advances $s\,\Delta t$
(no paper links the two clocks)

case (Klempt et al. 2026)	effect of s
6: one species wins	winner appears for $s \gtrsim 0.15$
3: coexistence	share moves by 0.03–0.06

2. cap $\phi_{\mathrm{cap}} = 0.9$



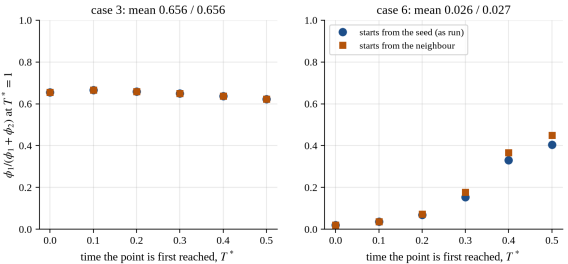
the amount given to the point model
is $\min(\phi_{3D}, \phi_{\mathrm{cap}})$

ϕ_{cap}	0.85	0.90	0.95
case 3 share	0.668	0.667	0.664

All other parameters are the published values.

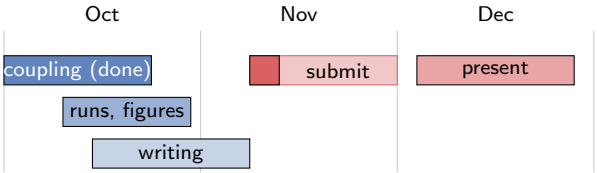
Limits

- composition is local, not carried with the front: effect on the mean ≤ 0.001 (right, Python)
- s has no source in a paper
- Eq. 34 as printed moves the colony instead of growing it (Appendix D)
- two α: the element's own and Eq. 36 are integrated separately (Check 2)



Composition at $T^* = 1$ of the points first reached at each time; Klempt 2024 test case 4.1 field (Python, 20 μm cube), Klempt 2026 cases 3 and 6.

Status and timeline



- examiners Prof. Junker (IKM), Dr. Wangenheim (IDS); supervisors Dr. Soleimani, Dr. Geisler
- draft sent on 9 October; **submission 13 November**
- colloquium 16 December, 13:00, IKM room 320
- growth from the living biomass: not in the thesis, continued at Keio

Questions for this meeting

1. **Your coupling ideas:** which direction would you take first?
2. **Growth law:** amount of the field (Eq. 36) or living biomass of the point model?
3. **Two-way coupling:** should the stress act back on the growth or the composition?
4. **Rate κ:** fixed by the mean growth of Eq. 36, or by the measured volume?
5. **Next steps:** what should be done here, and what at Keio?

References

- Klempt, Soleimani, Junker (2023), PAMM: precursor spatial model
- Klempt, Soleimani, Wriggers, Junker (2024), Biomech. Model. Mechanobiol.: growth field, Eq. 34/36, Table 2
- Klempt, Geisler, Soleimani, Junker (2026), Arch. Appl. Mech.: multi-species point model, Table 1 cases
- Fritsch et al. (2025), arXiv:2512.15145: Bayesian updating, same model

Appendix A: notation

Growth field and mechanics (Klempt et al. 2024)

ϕ	biofilm volume fraction (0 = void, 1 = dense)
c	nutrient concentration
α	expansion variable, $\mathbf{F}_g = \alpha \mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$
$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$	elastic / growth split
k_α	growth factor (10^{-3} per T^*)
β	phase-field regularisation
r, k	front growth rate, Monod constant
T^*	normalised time, $T^* \in [0, 1]$

Point model (Klempt et al. 2026)

ϕ_i	volume fraction of species i
ϕ_0	void, $\phi_0 = 1 - \sum_i \phi_i$
ψ_i	living fraction (viability) of species i
γ	Lagrange multiplier for $\sum \phi_i + \phi_0 = 1$
A_{ij}	interaction matrix (self, cooperation, competition)
b_i	sensitivity to the antibiotic
c^*, α^*	nutrient, antibiotic concentration
η_i	viscosity (time scale) of species i

Coupling (this work)

ϕ_{3D}	amount from the element field
$\phi_i / \sum_j \phi_j$	composition (share of species i)
s	time link: point model advances $s\,\Delta t$
ϕ_{cap}	largest amount the point model sees

Appendix B: background, growth field and mechanics

Kinematics. Growth is a volume change: $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$ with isotropic $\mathbf{F}_g = \alpha\mathbf{I}$. Stress comes only from $\mathbf{F}_e$ (neo-Hookean). A body that can grow freely stays stress free; stress appears where growth is constrained.

Material (Klempt et al. 2024, Eq. 20 and Table 2): $E = 10$ Pa, $\nu = 0.49$, stiffness weighted with ϕ^2 (void: floor $10^{-3}E$); no viscosity. Stresses scale linearly with E (the example input's 1000 MPa gives $10^8\times$ larger values).

Field equations (Klempt et al. 2024, from an extended Hamilton principle):

$$\dot{\phi} = \underbrace{\beta\,\Delta\phi}_{\text{diffusion}} + \underbrace{k_\alpha\,\alpha}_{\text{local source}} - \underbrace{r\,\frac{c}{k+c}\,\nabla\phi\cdot\frac{\nabla c}{|\nabla c|}}_{\text{front growth towards nutrient}}\,, \qquad \dot{\alpha} = k_\alpha\,\phi$$

- ϕ and c are nodal fields of the element; α is an internal variable at the Gauss points
- The front term moves the biofilm boundary up the nutrient gradient; it vanishes where ϕ or c is uniform
- The nutrient diffuses much faster than the biofilm grows (quasi-static c)

Appendix C: point model and one substep of the coupled run

Point model (Klempt et al. 2026), per species i, no space dependence, implicit, $\Delta t = 10^{-4}$:

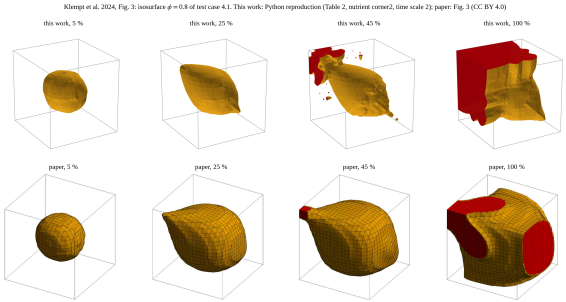
$$(\eta_{\phi,i} + \eta_i \psi_i^2)\,\dot{\phi}_i + \eta_i \phi_i \psi_i\,\dot{\psi}_i = c^* \psi_i \textstyle\sum_j A_{ij}\phi_j\psi_j - \gamma - (\text{barrier})$$

$$\eta_i\,(\phi_i \psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) = c^* \phi_i \textstyle\sum_j A_{ij}\phi_j\psi_j - \alpha^* b_i\,\psi_i - (\text{barrier}), \qquad \textstyle\sum_i \phi_i + \phi_0 = 1$$

One substep $t_n \to t_{n+1}$: ANSYS solves only $\operatorname{div}\boldsymbol{\sigma} = \mathbf{0}$; the material routine turns $\mathbf{F}$ into $\boldsymbol{\sigma}$ after removing the growth.

repeat (Newton iteration of ANSYS)
  $\mathbf{F}$ at every Gauss point from the current displacements
  for each Gauss point
    $\phi_{3D} = \phi_1 + \phi_2$ from the element; restore the converged state of t_n
    $\alpha_{n+1} = \alpha_n + k_\alpha\,\phi_{3D}\,\Delta t$ Eq. 36
    $\mathbf{F}_e = \mathbf{F}\,\mathbf{F}_g^{-1}$, $\mathbf{F}_g = \alpha_{n+1}\mathbf{I}$ $\to$ $\boldsymbol{\sigma}$, tangent
    if two species and $\phi_{3D} \geq \phi_{\min}$:
      scale the point model to $\min(\phi_{3D}, \phi_{\mathrm{cap}})$
      bridge (Fortran $\to$ C $\to$ socket $\to$ Python): n_{sub} steps of $s\Delta t/n_{\mathrm{sub}}$
      scale back; store ϕ_i, ψ_i, γ as state variables
    return $\boldsymbol{\sigma}$ and the tangent to ANSYS
  assemble the residual, update the displacements
until converged
the element advances ϕ_1, ϕ_2, c to t_{n+1} (its own solver)

Appendix D: Klempt et al. 2024 test cases, reproduced in Python



test case	time scale	RMS ϕ	RMS c
4.1 (Fig. 4)	2	0.03	0.02
4.2 “high” (Fig. 7)	10	0.04	0.04
4.2 “low” (Fig. 7)	4–5	0.09	0.05

Independent solver on the paper's 20 μm cube, not the ANSYS model. Table 2 unchanged; growth on both faces and consumption $g\phi c$ (two departures from Eq. 34/35 as printed); nutrient in a 2 μm corner block; one time scale per run ($T^* = t/t_{\mathrm{ref}}$, Sec. 4). Curves agree; the shape agrees in kind (egg towards the corner) but is more slender, also with other diffusion and a 2nd-order scheme. Table 3's pressure fits μ in MPa.

Appendix E: growth from the living biomass, implementation and runs

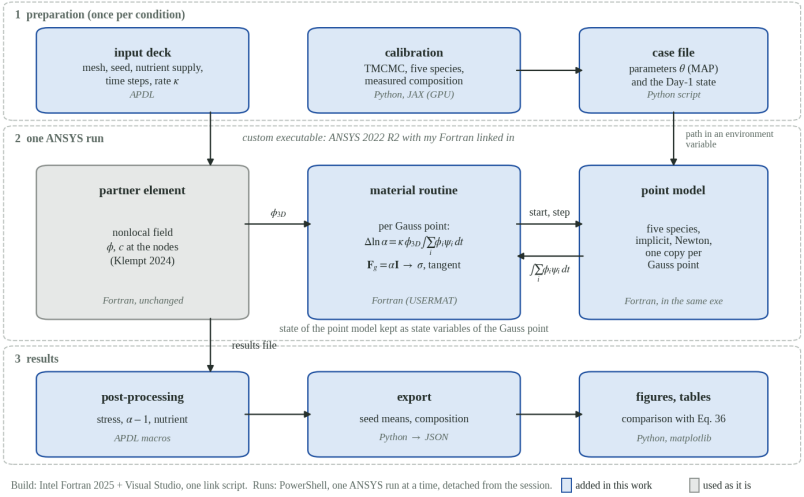
$$\ln \alpha_{n+1} = \ln \alpha_n + \kappa\, \phi_{3D} \int_{t_n}^{t_{n+1}} \sum_i \phi_i \psi_i \, dt \qquad \text{(Appendix C, line 5 only)}$$

item	how
form	increment of $\ln \alpha$, as Klempt 2024 Table 1
$\phi_{3D} \in [0, 1]$	only marks where the biofilm is
point model	one per Gauss point, on its own (not scaled to ϕ_{3D})
start	when $\phi_{3D} \geq \phi_{\min}$, from the calibrated Day-1 state
integral	over its own substeps, clock s per unit T^*
storage	state variables of the Gauss point; element unchanged
κ	$8.14 \cdot 10^{-3}$: mean $\alpha - 1$ as Eq. 36 ($-0.003\,\%$); assumed

seed, $T^* = 1$	Eq. 36	living	diff.
$\alpha - 1$, 8^3	$6.752 \cdot 10^{-4}$	$6.752 \cdot 10^{-4}$	0.0 %
von Mises/E, 8^3	$4.684 \cdot 10^{-5}$	$4.357 \cdot 10^{-5}$	$-7.0\,\%$
vM, 16^3; 24^3			-7.1; $-7.0\,\%$
vM/($E(\alpha - 1)$)	0.069	0.064	
antibiotic, 8^3	0 %	$-32\,\%$	

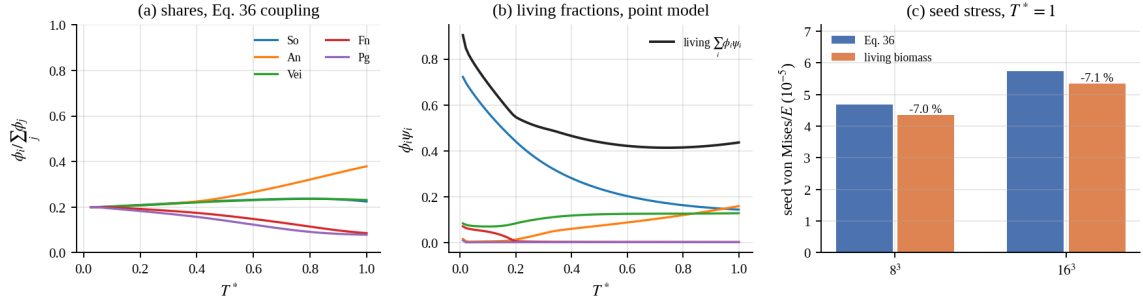
CH, $E = 10$ Pa, $\nu = 0.49$; the antibiotic case with the previous calibration. Same amount of growth, different spread inside the seed; $\alpha - 1$ and stress scale with κ.

Appendix F: how the run is implemented, and the tools



Blue: added in this work; grey: used as it is. One line of the material routine changes compared with Eq. 36.

Appendix G: calibrated five-species model in the element



	start	$T^* = 1$	growth $\alpha - 1$	seed von Mises
shares (Eq. 36)	0.2 each	An 0.38, Vei 0.23, So 0.23, Fn 0.09, Pg 0.08	$6.752 \cdot 10^{-4}$, same as two species	$\mathrm{vM}/E = 4.68 \cdot 10^{-5}$
living biomass $\sum_i \phi_i \psi_i$	0.90	0.44 (mostly So lost)	same mean	$-7.0\,\%$ (8^3), $-7.1\,\%$ (16^3)

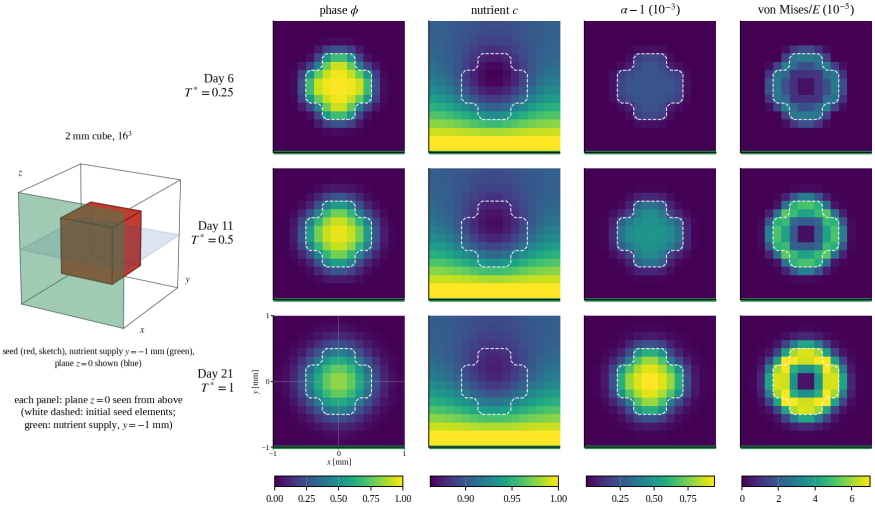
commensal flow condition (CH), calibration to the measured data, $E = 10$ Pa, 8^3

Appendix H: two and five species, implementation

	two species (5 October)	five species (since 10 October)
point model	Python, through a material server (Fortran $\rightarrow$ C $\rightarrow$ socket $\rightarrow$ Python)	Fortran, in the same executable (no server, about 4 times faster, same results)
parameters	Klempt et al. 2026 cases 3 and 6 ($c^* = 100$, $\alpha^* = 10$)	calibration (TMCMC, MAP, 5×5 interactions), $c^* = 25$, $\alpha^* = 0$
start	equal shares	equal shares; calibrated Day-1 state for the living-biomass growth
stiffness	partner's example input, $E = 1000$ MPa	Klempt 2024, $E = 10$ Pa
coupling (Eq. 36)	same code: amount scaled to ϕ_{3D}, $\phi_{\mathrm{cap}} = 0.9$, time link $s = 0.15$	
growth from the living biomass	–	yes; point model runs on its own, $s = 0.2375$ ($T^* = 1$ is Day 21)

The number of species is a parameter (1 to 5) of the same routine.

Appendix I: fields on the mid-plane over time



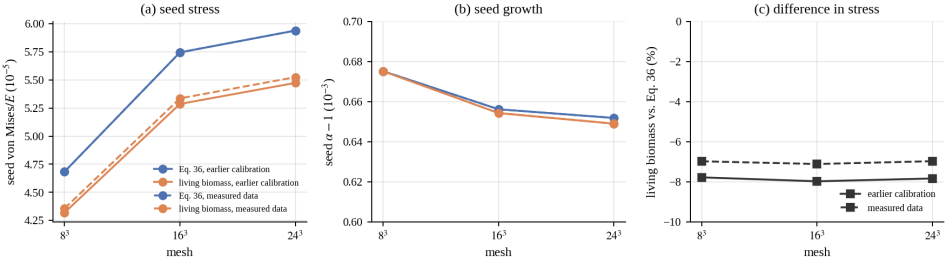
Plane $z = 0$ seen from above (x right, y up), 16^3, CH calibrated to the measured data. Growth by Eq. 36. The phase spreads and thins, the nutrient is used around the seed, and the stress forms a ring at the seed edge.

Appendix J: analysis conditions

geometry, mesh	2 mm cube; 8^3 (512), 16^3, 24^3 hexahedral elements; seed of 32 elements at the centre (8^3, refined with the mesh)
mechanical BC	only rigid-body motion fixed (corner nodes); otherwise traction free
nutrient BC	$c = 1$ on the face $y = -1$ mm; no flux elsewhere
initial state	$\phi = 1$ in the seed, ≈ 0 outside; $\alpha = 1$; $c = 1$ everywhere
time	$T^* = 1$ is Day 21; $\Delta T^* = 0.025$ (40 steps; living-biomass runs: 1/95); ANSYS 2022 R2, static, nonlinear geometry
material	neo-Hooke, $E = 10$ Pa, $\nu = 0.49$, $E \propto \phi^2$ (Klempt 2024 Table 2); $\mathbf{F}_g = \alpha \mathbf{I}$
growth field	$k_\alpha = 10^{-3}$, $\beta = 0.02$ mm$^2/T^*$ (Table 2 scaled to the 2 mm cube, assumed)
point model	two species: Klempt 2026 cases 3 and 6; five species: calibrated MAP (CH), $c^* = 25$
coupling	$\phi_{\mathrm{cap}} = 0.9$, $\phi_{\mathrm{min}} = 0.01$, time link $s = 0.15$ (assumed); living-biomass growth: $s = 0.2375$, $\kappa = 8.14 \cdot 10^{-3}$ (assumed)
output	seed means of $\alpha - 1$ and von Mises$/E$ at $T^* = 1$

Values not from a paper are marked as assumed.

Appendix K: mesh convergence



mesh	elements	nodes	integration points	seed elements
8^3	512	729	4 096	32
16^3	4 096	4 913	32 768	256
24^3	13 824	15 625	110 592	864

8 integration points per hexahedral element ($2 \times 2 \times 2$)

- seed stress: $+23\,\%$ from 8^3 to 16^3, $+3.4\,\%$ from 16^3 to 24^3; converging
- seed growth changes by less than $3.5\,\%$
- difference of the growth laws: -7 to $-8\,\%$ on every mesh
- measured-data calibration: $-7.0\,\%$ at 8^3 and 24^3, $-7.1\,\%$ at 16^3

Appendix L: finite element formulation

balance	$\operatorname{div}\boldsymbol{\sigma} = \mathbf{0}$ in the current configuration, no body force; weak form by ANSYS (static, large deformation)
kinematics	$\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$, $\mathbf{F}_g = \alpha\mathbf{I}$; Cauchy stress from $\mathbf{F}_e$
material	neo-Hooke in $\mathbf{F}_e$, stiffness $(\phi^2 + f)E$ with a small void part f; tangent returned by the material routine
element	SOLID185, 8 nodes, $2 \times 2 \times 2$ integration with B-bar (default), for $\nu = 0.49$
nonlinear solver	full Newton–Raphson, fixed steps (no automatic time stepping), up to 50 iterations, default force tolerance
coupling	staggered: ANSYS solves the mechanics with ϕ fixed; at the end of each step the partner element updates ϕ explicitly and solves the nutrient (linear system, PARDISO) on its neighbour operator
stability	explicit update of ϕ: $\Delta t \leq h^2/(2\beta)$; met for all meshes used
supports	3-2-1: one corner node u_x, u_y, u_z, one u_y, u_z, one u_z (6 dofs, only rigid-body motion)

Time step (one species, 8^3, $T^* = 1.1$): $\Delta T^* = 0.1 \to 0.05 \to 0.025$ changes the seed von Mises/E by $-3.0\,\%$ and $-1.5\,\%$ ($\alpha - 1$: $-0.7\,\%$, $-0.3\,\%$), first order; the runs use $\Delta T^* = 0.025$, the living-biomass runs $1/95 \approx 0.011$.